SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA1105

COURSE OUTCOME COVERED

CO2: Design UML diagrams to represent system requirements and architecture. *(BL3)*

Assessment Tool Weightage: 40%

QUESTIONS: Any 4 **Total Marks:40**

Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

I DARSHINIE KARANI.V.M (192520033) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

ANALYTICAL PROBLEM SOLVING

1. Use Case Diagram Problem

A **Smart Banking System** allows customers to create accounts, transfer funds, pay bills, and view transaction history. Bank employees verify customer details, while administrators manage user accounts and banking services. The system requirements are only partially documented.

Tasks:

a) Identify all actors involved in the system.
b) List the primary use cases.
c) Define relationships (include, extend, generalization).
d) Construct a complete **Use Case Diagram** representing system functionality.

(10 Marks)

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2. Class Diagram Problem

An **Online Vehicle Rental System** consists of entities such as Vehicle, Customer, Reservation, Payment, and RentalAgent. The relationships among these entities have not been clearly defined.

Tasks:

a) Identify classes with appropriate attributes and methods.
b) Define relationships (association, aggregation, inheritance, composition where applicable).
c) Specify multiplicity between classes.
d) Design a detailed **Class Diagram** for the system.

(10 Marks)

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3. Sequence Diagram Problem

An **E-Commerce Shopping System** allows a customer to browse products, add items to the cart, place an order, make payment, and receive order confirmation. The sequence of interactions between system components is unclear.

Tasks:

a) Identify all objects participating in the interaction.
b) Define the message flow between objects.
c) Represent activation bars and lifelines.
d) Construct a **Sequence Diagram** showing the complete interaction flow.

(10 Marks)

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4. Activity Diagram Problem

A **Job Recruitment Portal** enables applicants to register, upload resumes, apply for jobs, attend interviews, and receive offer letters. Recruiters verify applications before scheduling interviews. The business workflow needs optimization.

Tasks:

a) Identify all activities in the recruitment workflow.

b) Add decision nodes for selection and rejection.
c) Include parallel activities wherever applicable.
d) Design an optimized **Activity Diagram** representing the complete workflow.
(10 Marks)

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5. State Diagram Problem

A **Smart Parking Management System** operates through states such as Parking Slot Available, Vehicle Entered, Parking Allocated, Payment Pending, Payment Completed, and Vehicle Exited. The system behavior during parking operations needs to be modeled.
Tasks:
a) Identify all possible states of the system.
b) Define events triggering state transitions.
c) Include initial and final states.
d) Construct a **State Transition Diagram** representing system behavior.
(10 Marks)

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6. Deployment Diagram Problem

A **Smart Agriculture Monitoring System** uses IoT sensors to monitor soil moisture, temperature, and humidity. Sensor data is transmitted to a cloud platform where farmers monitor crop conditions through a mobile application. Automated irrigation is activated whenever soil moisture falls below a predefined threshold.
Tasks:
a) Identify all hardware nodes involved in the system.
b) Specify software components deployed on each node.
c) Define communication links between hardware nodes.
d) Construct a complete **Deployment Diagram** representing the system architecture.
(10 Marks)
RUBRICS FOR CALCULATION

ANALYTICAL PROBLEM SOLVING

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

ANSWERS

1.

a) Actors:

1. Customer
    - Creates a new bank account.
    - Transfers funds.
    - Pays utility and other bills.
    - Views transaction history.
2. Bank Employee
    - Verifies customer identity and documents.
    - Approves account creation.
3. Administrator
    - Manages user accounts.
    - Manages banking services.
    - Maintains the overall system.

b) The primary use cases of the Smart Banking System are:

- Create Account
- Verify Customer Details
- Transfer Funds
- Pay Bills
- View Transaction History
- Manage User Accounts
- Manage Banking Services
- Send Transaction Notification

c) <<include>> Relationship

- Create Account <<include>> Verify Customer Details
- Verification is compulsory before creating a bank account.

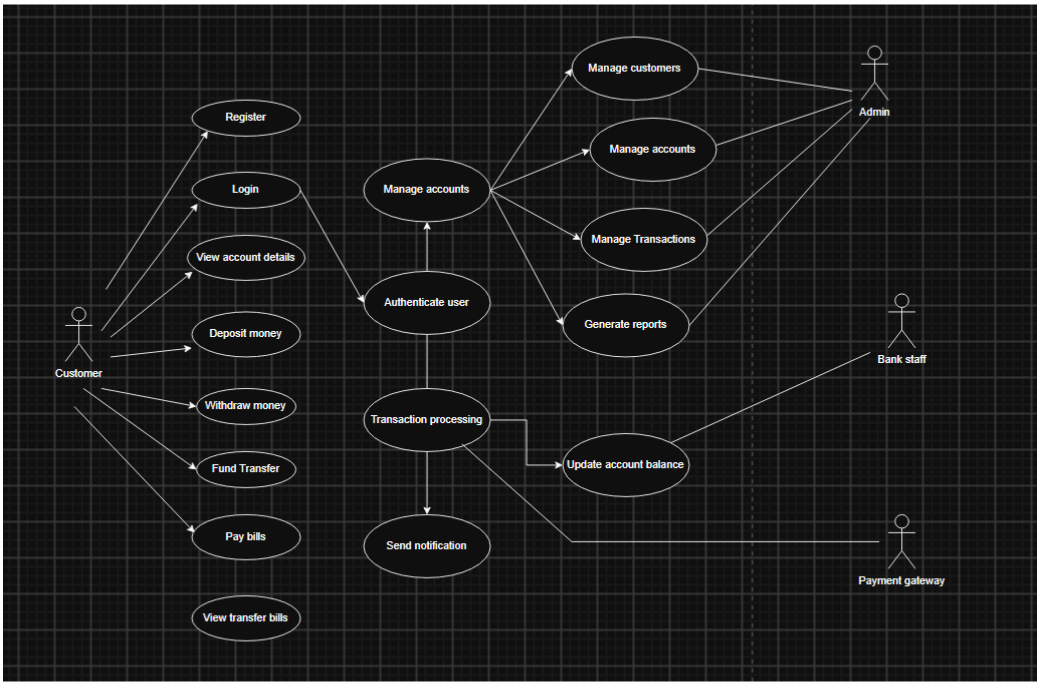
<<extend>> Relationship

- Transfer Funds <<extend>> Send Transaction Notification
- After a successful fund transfer, the system sends an SMS or email notification.

Generalization Relationship

- Administrator is a specialized Bank Employee, as the administrator performs all employee functions along with additional administrative responsibilities.

d)



2.

1. Vehicle

Attributes

- vehicleID
- vehicleName
- model
- rentalPrice
- availabilityStatus

Methods

- checkAvailability()
- updateStatus()

2. Customer

Attributes

- customerID
- customerName
- phoneNumber
- address

Methods

- register()

- bookVehicle()
- cancelReservation()

3. Reservation

Attributes

- reservationID
- bookingDate
- rentalDuration
- reservationStatus

Methods

- createReservation()
- confirmReservation()
- cancelReservation()

4. Payment

Attributes

- paymentID
- amount
- paymentMethod
- paymentStatus

Methods

- makePayment()
- generateReceipt()

5. RentalAgent

Attributes

- agentID
- agentName
- contactNumber

Methods

- verifyCustomer()
- approveReservation()

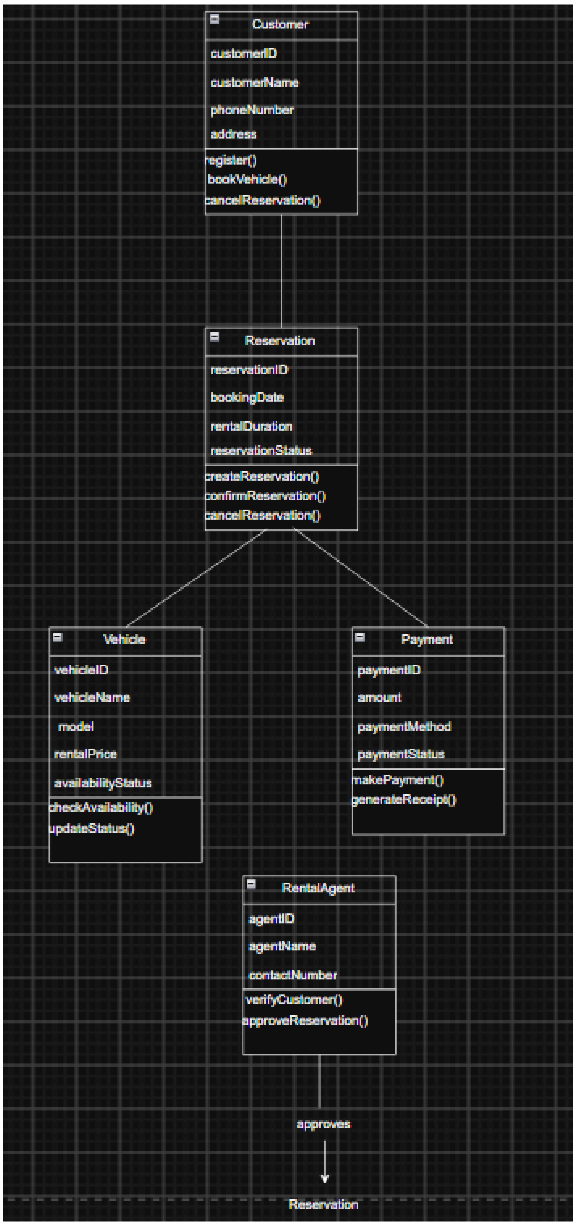
b) Association: Customer books a Reservation.

- Association: Reservation is made for one Vehicle.
- Composition: Reservation contains Payment. (Payment cannot exist without Reservation.)
- Association: RentalAgent approves Reservation.
- Inheritance: Administrator can inherit from RentalAgent if an admin role is included.

c)

Relationship	Multiplicity
Customer → Reservation	1 : 0..*
Reservation → Vehicle	1 : 1
Reservation → Payment	1 : 1
RentalAgent → Reservation	1 : 0..*

d)



3.

a) The objects involved in the E-Commerce Shopping System are:
1. Customer
2. Product Catalog
3. Shopping Cart
4. Order
5. Payment Gateway
6. Order Confirmation System

b)

The sequence of messages is:

1. Customer browses products.
2. Product Catalog displays available products.

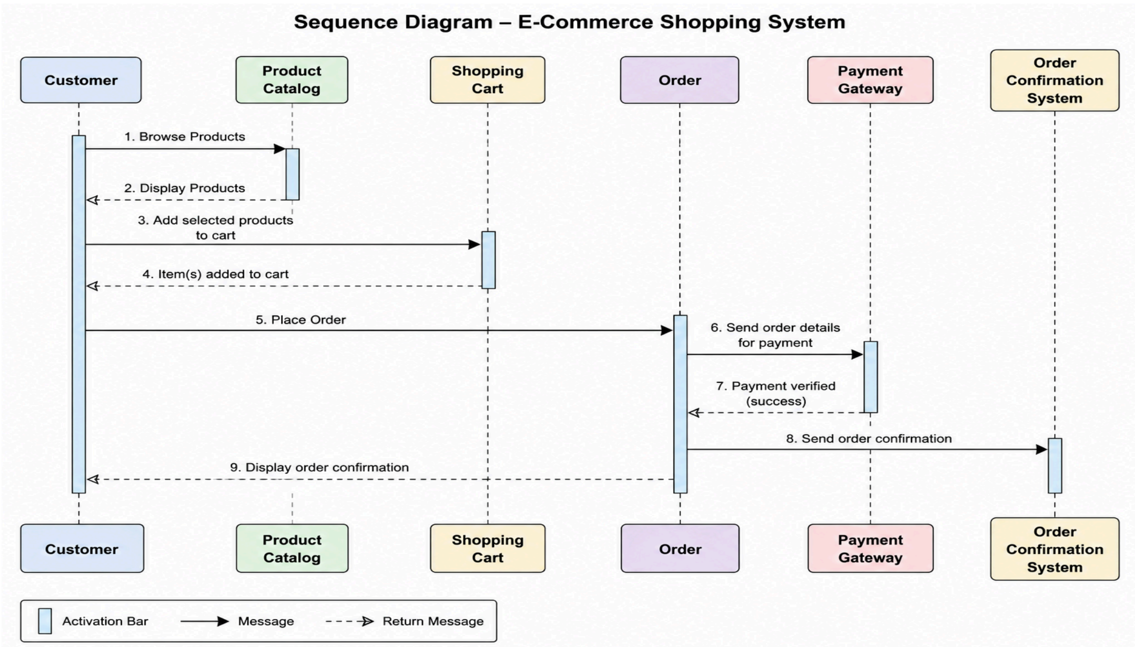
3. Customer adds selected products to the Shopping Cart.
4. Customer places an order.
5. Order details are sent to the Payment Gateway.
6. Payment Gateway verifies and processes the payment.
7. Payment success message is sent to the Order System.
8. Order Confirmation System generates and sends the confirmation to the Customer.

c)

Lifeline: Represents the existence of each object during the interaction.

Activation Bar: Represents the time during which an object is performing an operation after receiving a message.

d)



4.

a) The activities involved in the Job Recruitment Portal are:

1. Applicant Registration
2. Login
3. Upload Resume
4. Search for Jobs
5. Apply for Job
6. Recruiter Verifies Application
7. Schedule Interview
8. Attend Interview
9. Selection Decision
10. Receive Offer Letter
11. Reject Application

b) The activity diagram includes two decision points:

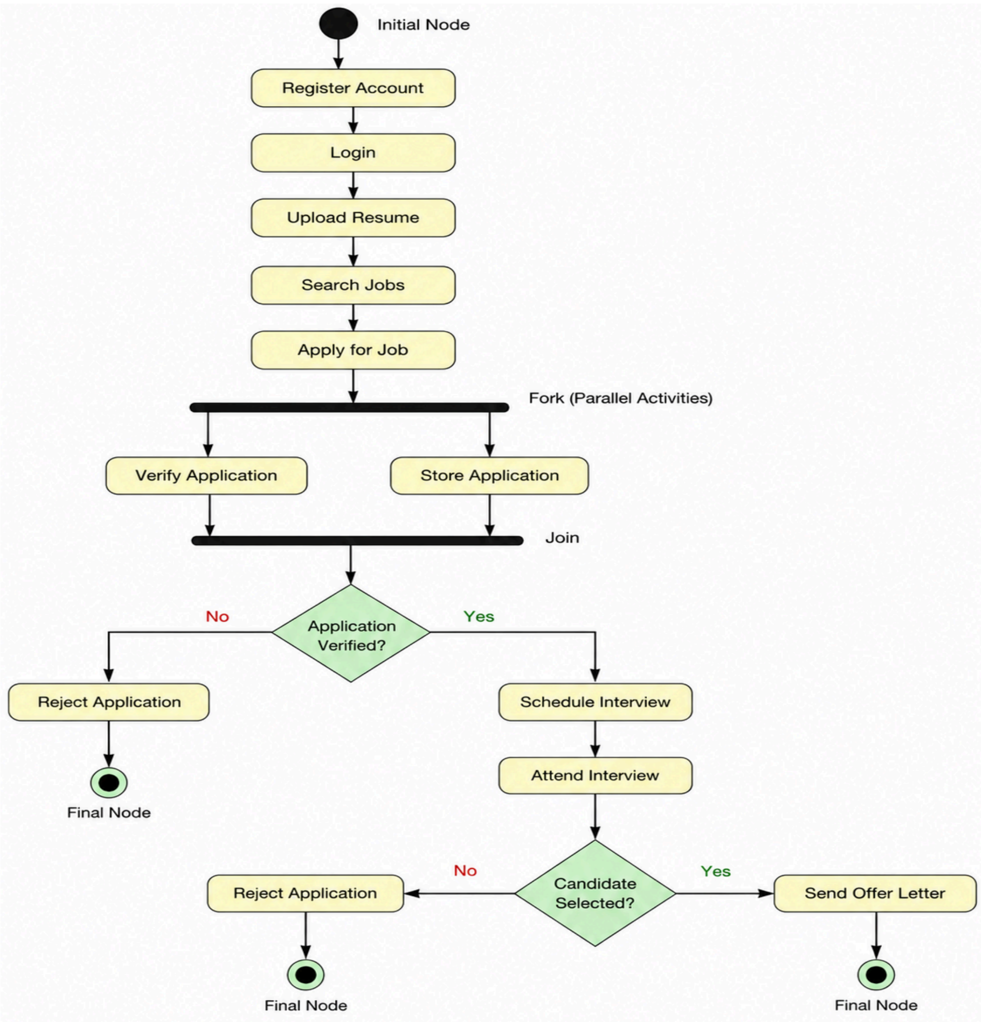
1. Application Verified?
   - Yes → Schedule Interview
   - No → Reject Application
2. Candidate Selected?
   - Yes → Send Offer Letter
   - No → Reject Application

c) Parallel activities in the recruitment process are:

- After the applicant submits the application:
  - The Recruiter verifies the application.
  - The System stores the application details in the database simultaneously.

These activities occur in parallel before proceeding to the interview stage.

d)



5.

a) The Smart Parking Management System passes through the following states:

1. Initial State
2. Parking Slot Available
3. Vehicle Entered
4. Parking Allocated
5. Payment Pending
6. Payment Completed
7. Vehicle Exited
8. Final State

b)

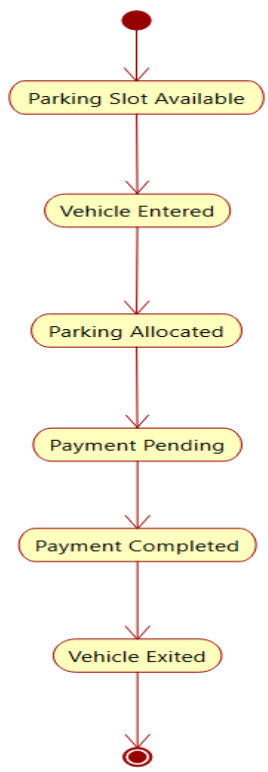
Current State	Event	Next State
Initial State	Parking slot available	Parking Slot Available
Parking Slot Available	Vehicle enters	Vehicle Entered
Vehicle Entered	Slot allocated	Parking Allocated
Parking Allocated	Parking session ends	Payment Pending
Payment Pending	Payment successful	Payment Completed
Payment Completed	Vehicle exits	Vehicle Exited
Vehicle Exited	Exit completed	Final State

c)

Initial State: The parking system starts and waits for an available parking slot.

Final State: The parking process ends after the vehicle exits successfully.

d)



6.

a) The hardware nodes in the Smart Agriculture Monitoring System are:

1. IoT Sensors (Soil Moisture, Temperature, Humidity Sensors)
2. Gateway Device (Wi-Fi/IoT Gateway)
3. Cloud Server
4. Farmer's Mobile Device
5. Irrigation Controller
6. Water Pump/Sprinkler System

b)

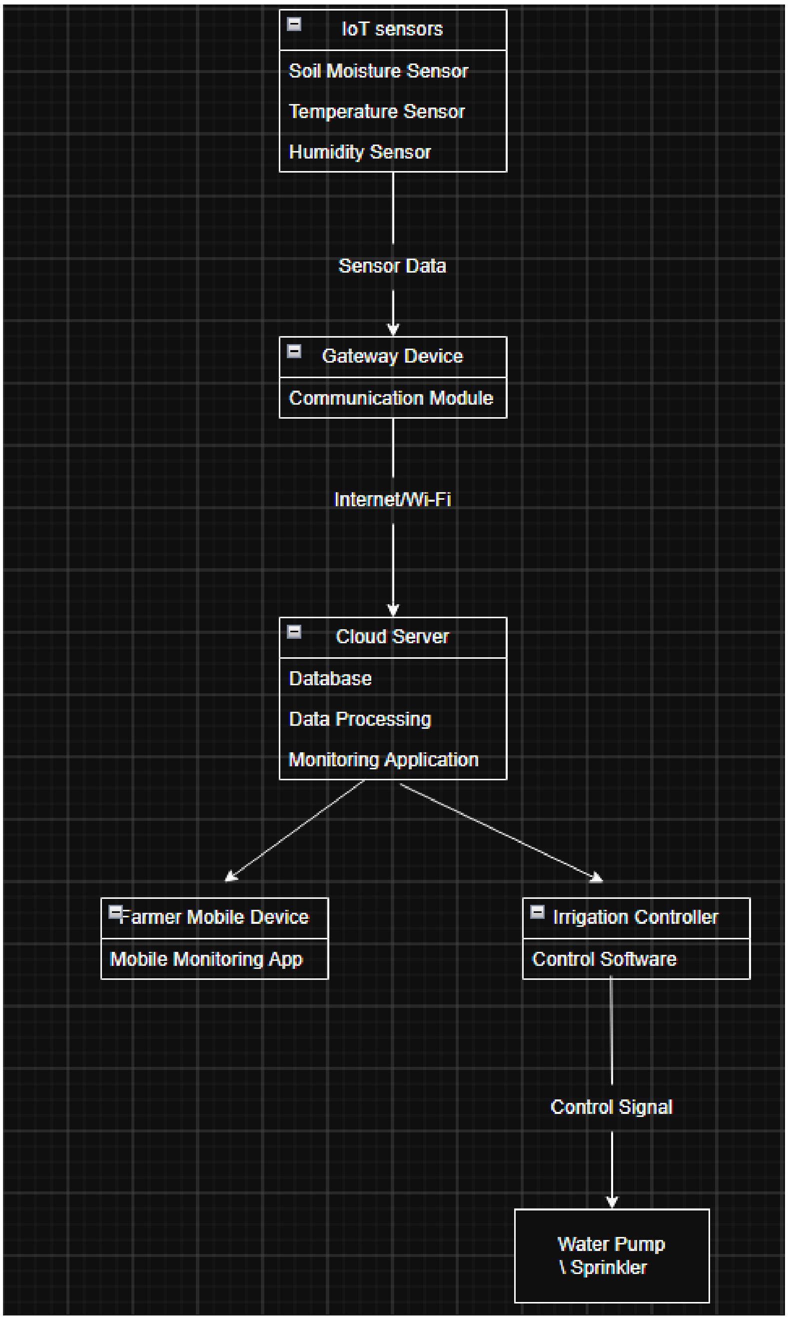
Hardware Node	Software Component
IoT Sensors	Sensor Firmware
Gateway Device	Data Collection & Communication Module
Cloud Server	Database, Data Processing, Monitoring Application

Mobile Device	Farmer Mobile App
Irrigation Controller	Irrigation Control Software

c) IoT Sensors → Gateway Device: Sensor data (soil moisture, temperature, humidity).

- Gateway Device → Cloud Server: Internet/Wi-Fi communication.
- Cloud Server → Mobile App: Displays crop and field monitoring data.
- Cloud Server → Irrigation Controller: Sends irrigation commands.
- Irrigation Controller → Water Pump: Activates irrigation when soil moisture is below the threshold.

d)


IoT sensors
Soil Moisture Sensor
Temperature Sensor
Humidity Sensor
Sensor Data
Gateway Device
Communication Module
Internet/Wi-Fi
Cloud Server
Database
Data Processing
Monitoring Application
Farmer Mobile Device
Mobile Monitoring App
Irrigation Controller
Control Software
Control Signal
Water Pump
\ Sprinkler